

Hiver AI Customer Support Agent for AppleSupport

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Submission For: Hiver SDE Intern Take-Home Assignment | **Repository:**
https://github.com/Anuragpathak07/Hiver_Submission.git

Executive Summary & Headline Evaluation Metrics

Evaluation Metric	System Result	Benchmark Baseline	Production Significance
Intent Classification Accuracy	85.00%	15.50% (Majority) 79.50% (TF-IDF + LogReg)	+5.50% improvement over strong TF-IDF baseline across 8 intents.
Intent Classification Macro-F1	0.8512	0.0335 (Majority) 0.7954 (TF-IDF + LogReg)	High balanced performance with zero class collapse.
Historical Retrieval Match @ 3	87.50%	69.50% (Lexical TF-IDF)	+18.00% improvement using semantic vector embeddings.
Retrieval Top-1 Avg Similarity	0.7142	0.3711 (Lexical TF-IDF)	Captures technical symptom paraphrasing accurately.
LLM-as-Judge Quality Score	4.25 / 5.0	80.00% Rubric Pass	Evaluated across 6 quality criteria (correctness, grounding, tone).
Human-Judge Agreement	90.00%	30-case validation subset	Strong alignment between automated LLM judge and human evaluation.
Auto-Handle Rate (AI Agent)	73.00%	146 / 200 Golden cases	Safely resolves routine diagnostic & update queries.
Escalation Rate (Human Specialist)	27.00%	54 / 200 Golden cases	Safely escalates high-risk identity, billing, or low-confidence queries.

1. Problem Framing & Brand Selection

1.1 Brand Selection Rationale: We selected **AppleSupport** from Kaggle's 3M-tweet dataset (~106,000+ tweets) due to its high multi-turn diagnostic depth (asking for iOS versions, storage, SMC resets), recurring technical issue volume (iOS 11 & macOS High Sierra releases), and clear ground-truth support resolutions.

1.2 What 'Good' Means for AppleSupport:

- **Grounded Precision:** Responses must be strictly backed by historical resolutions without inventing steps or dead links.
- **Conservative Escalation:** High-risk queries (account recovery, billing disputes) or low-confidence predictions (<0.50) must be escalated to human agents with a clear diagnostic reason.
- **Reproducible Proof:** Offline evaluation must demonstrate measurable superiority over trivial and simple baselines on a clean benchmark.

What We Chose Not to Build: Automated live Twitter posting (agent generates drafts for human review), multi-lingual translation layers, and multi-modal image screenshot parsing.

2. Dataset Journey & Intent Discovery

Raw tweets are flat and noisy. We parsed `in_response_to_tweet_id` recursively to reconstruct full conversation trees, producing **80,672 complete AppleSupport cases**. Using `sentence-transformers/all-MiniLM-L6-v2` embeddings + KMeans ($k=12$), followed by Groq (`openai/gpt-oss-20b`) JSON Schema consolidation, we defined an **8-intent taxonomy**: `ios_update_issues`, `battery_life_issue`, `macbook_issues`, `keyboard_autocorrect_issue`, `music_audio_issue`, `iphone_hardware_issue`, `account_identity_issue`, and `other_unclear`.

3. Hand-Annotation & Golden Set Methodology

We built a **200-example Golden Evaluation Set** using stratified sampling (~25 cases/intent, fixed seed=42). To prevent data leakage, all 200 Golden Set case IDs were **strictly excluded** from classifier training data and vector indices. Evaluators verified each candidate case using an interactive CLI tool (`src/evaluation/annotate_golden.py`). Silver vs. Human agreement reached **92.50%** (185/200), and 15 human-corrected disagreements were frozen at `data/golden/golden_annotated.csv`.

4. Results vs. Two Baselines

- **Baseline 1 (Trivial Majority Class):** Always predicts `ios_update_issues`. Accuracy = **15.50%**, Macro-F1 = **0.0335**.
- **Baseline 2 (Simple TF-IDF + LogReg):** Fits TF-IDF (10k features) + LogisticRegression($C=1.0$). Accuracy = **79.50%**, Macro-F1 = **0.7954**.
- **Our Model (SentenceTransformer + LogReg):** Dense embeddings + Calibrated LogReg($C=2.0$). Accuracy = **85.00%**, Macro-F1 = **0.8512**.
- **Retrieval System (Semantic vs Lexical):** Semantic Vector Search achieved **87.50% Match@3** (0.7142 Top-1 Sim) vs **69.50%** for Lexical TF-IDF.

5. Top 5 Real Failure Modes Analysis

1. **Hardware vs. Software Symptom Ambiguity:** Update-induced hardware symptoms (e.g. contacts app freeze post-update) misclassified as hardware (`apple_2750532`).
2. **Short / High-Noise Customer Tweets:** Expletive/short queries (e.g. *'what the fuck'*) yield low retrieval similarity (safely caught by Escalation Engine).
3. **Taxonomy Boundary Blur:** Purchasing questions overlapping between `account_identity_issue` and `other_unclear`.
4. **Retrieval Response Divergence:** High query match where historical support response was a generic DM request.
5. **Emoji/Slang Over-conservatism:** Informal syntax lowering classifier confidence score below 0.50.

6. Mandatory Section: 'What is Misleading About My Headline Number?'

While our headline numbers (**85.0% Accuracy**, **87.5% Retrieval Match**, **4.25/5.0 Quality Score**) demonstrate a strong system, presenting them without context is misleading: (1) At $N=200$, the 95% confidence interval spans **[80.0%, 90.0%]** (margin of error $\pm 5.0\%$); (2) Stratified Golden Set sampling overrepresents rare classes compared to real Twitter volume where update issues comprise $>55\%$; (3) Pre-annotating silver labels introduced mild confirmation bias (92.5% agreement); (4) 2017 iOS 11 model training faces temporal distribution shift on modern iOS 17/18 queries; and (5) Intent Match @ 3 measures category overlap rather than end-user problem resolution.

7. Next-Week Plan & Key Engineering Decision Log

Next Week Plan: Blind double-annotation on N=1,000 cases, raw t.co link canonicalization to live Apple support URLs, contrastive fine-tuning on query-response pairs, and LLM response filtering.

Decision Log Highlights (12 Decisions): Selected AppleSupport for multi-turn depth; reconstructed 80,672 cases from parent chains; discovered 8 intents via embeddings + Groq consolidation; enforced zero-leakage Golden Set isolation; built dual semantic vector retrieval; implemented deterministic multi-tiered escalation; and configured reproducible <5 min offline evaluation execution.